

UNIT 3 Loading the MS Marina

Cranes on Marina: Lifting capacity: 5-60 tons

Radius: 12-40 metres

The cranes have a low height. This improves visibility from the vessel's navigating bridge. The cranes are light and have a low centre of gravity. This improves the ship's stability and increases cargo capacity.

The cranes have a wide rope field at the hook. This allows safe and stable cargo handling (anti-swing device). They have maintenance-free hydraulic luffing arms. The boom is well supported at all boom angles due to the double acting hydraulic rams.

The jib is above the operator's cabin. This gives the crane driver a free view of the ship deck and the cargo holds. The jib has a short minimum radius. This makes it possible to place cargo quite near the crane.

Down on the quay the stevedores are getting ready to load the Marina for her next voyage. She has her own loading gear on board; there are three huge Liebherr el/hydraulic cranes designed for multi-purpose cargo and container handling.

John Andersen, the Chief Mate is on the bridge supervising loading. He has radio contact with the crane drivers and the seamen working with the cargo down in the holds. He is also in contact with Timo Ranta, the Second Mate, who is supervising loading work on deck.

- Prepare the vessel for loading! Unlock the hatch covers! Timo, I'm giving notice to the stevedores on the quay that we expect to be ready to start loading by 14.00 local time. Is that enough time for preparations?
- Yes, that should be OK, John. Is the cargo list available and complete?
- Yes, the list is complete. Have you got the stowage plan?
- Yes, I have. It is a bit unclear for holds 4 and 5.
- All right, complete the stowage plan and report back to me, please! You have the stability calculations?

After a while Timo reports back.

- Stowage plan now complete. All is under control, John. Shall I prepare holds and hatches for loading now?
- Yes, prepare holds and hatches for loading, carry on.

Timo is now giving orders to the seamen working in the holds:

- Open the hatches for loading. Check the hatch covers for damage and report!
- Hatch covers in order!
- Switch on hold lights and hold ventilation.
- Hold lights switched on, hold ventilation system operational.
- Ok, are the holds clean and free of smell?
- Yes, holds clean and free of smell.
- Are the safety arrangements in the holds operational?
- Safety arrangements will be operational in 15 minutes.
- Everything clear down here John!
- OK, are the cranes operational?
- Yes, cranes are operational.
- Good. Instruct the crane drivers and keep within the safe working load of the cranes. Report back to me when loading is complete. We must check for possible damage and prepare for sea. That's all for now. Carry on!

While John is waiting he decides to have a much needed cup of coffee. The day goes on...

General stowage

The Chief Mate is generally responsible for loading and discharging of the vessel. All cargo handling gear on board needs frequent overhauling. If wires, blocks, chains, shackles, cargo hoses, hooks and slings are not checked regularly, loading and discharging can become a risky affair or cause unnecessary delay.

Carrying general cargo and its many separate items in drums, barrels, crates, bags, rolls, cases, boxes and cartons often presents difficulty. Check that this packing is not damaged or deformed. When loading general cargo, take special care not to waste

loading space. Take into account the nature of the cargo. Some merchandise gives off a strong pungent smell or moisture and is liable to contaminate other goods. Cover contaminated goods with tarpaulins and separate them from other goods. Handle fragile goods with caution. Do not stow heavy pieces of cargo on fragile ones; barrels of liquids may leak and cause damage etc.

Container damage: Segregation, Infestation

Today most general cargo is transported in containers. Before stuffing, it is important that containers are checked for external damage such as holes or tears in the panelling or broken or distorted or deformed door hinges, locks or door seals. There must be no internal damage such as leftovers (dust, grease or liquid) of the previous cargo and the interior should be free from any sweat or frost. There should be no evidence of pests, insects, vermin or rodents which might infest the cargo and lead to delay by the Port authorities.

Other rules concerning stowing goods into containers include the following:

- Do not load goods with damaged or deformed packing into the containers.
- Do not stow wet and damp goods with dry goods.
- Do not use dunnage or packaging which is incompatible with the cargo.
- Do not stow goods with tainting odours close to sensitive merchandise.
- Observe all IMO regulations concerning dangerous goods and use appropriate labels on the containers to identify such cargo.
- Comply with IMO stowage segregation at all times.
- Be sure to instruct the crew on how to use equipment, e.g. how to connect reefer plugs and clip-on units.

Ventilation

In cargo handling the Mate's knowledge of proper ventilation on board is of great importance. In loading and discharging as well as in making repairs on board, the crew members or stevedores may be poisoned or even killed by contaminated air or poisonous gases due to lack of ventilation.

Before anybody enters oil tanks, ballast tanks, storerooms, holds or pump-rooms to carry out repairs or inspections, those spaces must be properly ventilated. Without breathing apparatus, protective clothing or a smoke mask and a lifeline men risk their

lives if the spaces are not gas free. There have been examples of victims of suffocation on board ships due to accumulation of lethal gases or lack of oxygen.

Humidity, Condensation, Sweating

Container or cargo sweating occurs when the outer surface of the container/cargo hold is cooled to a temperature below that of the air enclosed in the container/hold. This results in condensation: droplets forming on the interior roof and interior side panels, and then dripping on to the cargo, causing mould and water damage.

This might happen, for example when cargo loaded in the tropical belt in warm conditions with high relative humidity is transported to cold winter conditions in Europe. The temperature outside the container gradually cools down during transit, leading to condensation inside the container. One simple solution may be to replace the warm moist air by proper ventilation or air conditioning.

Sometimes it is important that the air surrounding a cargo has to be very dry. Ventilated air then has to be passed through dehumidifier units. Various desiccants and absorbent materials can also be used to absorb moisture and help keep cargo dry.

Dangerous Cargo

Today a great deal of the cargo carried on board ships falls under the category of dangerous goods. Knowledge of the IMO classification of dangerous goods is vital and it is equally important always to observe the IMDG Code when handling such goods. Explosives, flammable substances, poisonous or noxious cargo, infectious substances, radioactive materials and corrosives all require special precautions as to handling, stowing, segregation and labelling. So do goods that are liable to spontaneous heating and combustion.

Always check the compatibility of IMO class goods and ensure proper segregation. Thus, for example, flammable cargo must be stowed away from the engine room. Infectious substances must be separated by one hold/compartment from foodstuffs.

Spillage

Cleanliness in the engine room is important in order to prevent excessive oil residue in bilges and throughout the engine room. Drip trays, oil pumps, fuel oil valves and sea suction valves must be in good working condition.

Spills of fuel oil on board should end up in the sludge tank but today's mixtures of fuel oils, resulting from drainage and leakage in machinery spaces, lubricants, detergents, solvents and water often find their way into the bilge water tanks. Without modern bilge water cleaning equipment this may lead to discharge of bilge water containing pollutants at sea.

Ballasting

Ships have to de-ballast as cargo is loaded and ballast as cargo is off-loaded. Ballasting is a great problem from the point of view of pollution. "Tankers carry their cargo in a number of tanks or compartments within the hull of the ship. Before the introduction of segregated ballast tanks, tanks were cleaned after the oil was discharged and about one third of them filled with [ballasting] seawater... This naturally led to a considerable amount of oil getting into the sea. The ballast water, which was pumped overboard to make way for a fresh cargo of oil, was also contaminated."¹

"In the late 1970s an improvement was introduced. Instead of using water, the tank cleaning machines used crude oil - in other words, the cargo itself. When sprayed onto the sediments clinging to the tank walls, stripping the tanks, the oil simply dissolved them, turning them back into usable oil that could be pumped off with the rest of the cargo. There was no need for slop tanks to be used since the process left virtually no slops. The process became known as Crude Oil Washing (COW)"².

¹ International Maritime Organization (IMO). 2002. *Crude Oil Washing* [online].

Available: http://www.imo.org/Newsroom/mainframe.asp?topic_id=306

² International Maritime Organization (IMO). 2002. *Crude Oil Washing* [online].

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